

Proxmox Storage DRS -- Internals

How the tool works, for whoever changes it

Bernd Zeimetz bernd@bzed.de

2026-09-06

Rendered from docs/internals/*.md, SHA-256 7c5f484e09fc1553700687539f81b86942d683edeaa9abd7c004b751107d494
(sha256sum --check docs/internals.pdf.sha256 in the repository must succeed).

Copyright © 2026 Bernd Zeimetz. Licensed under the GNU Affero General Public License, version 3 or later.

Contents

The pipeline, as specified and as built	5
Module layout, as it stands	6
Why config.py depends on forecast.py	7
Where to read next	7
How a YAML file becomes a validated Config	7
Resolution order	8
The two-stage validation	8
Secrets from the environment	8
What is deliberately <i>not</i> checked here	8
ResolvedConfig: what the caller actually gets	9
Persistent state: state.json	9
One dataclass tree, matching section 11.2's JSON exactly	9
Reading degrades; writing does not	9
Locking: flock() is the mechanism, the JSON lock field is a label	9
What today's wiring into cli.py actually changes	10
Cooldowns: read by topology.py and heuristic.py, not by this module	10
Deliberately not implemented in this pass	10
Forecasting: the protocol and its three implementations	11
The one rule, in one place	11
Why the upper bound, never the point estimate	11
QuantileForecaster	11
SeasonalNaiveForecaster	11
HoltWintersForecaster	12
build_forecaster()	12
The Prometheus client and verify-metrics	12
_SessionLike / _ResponseLike: why not requests.Session directly	12
_get(): one request path, three endpoint shapes	12
PromQL construction	12
verify_metrics(): six checks, six functions	13
What cli.py does with the report	13
Command dispatch, the mode-override rule, and why logs go to stderr	13
One parser, global options first	13
Command handlers: a dict, not a chain of ifs	14
--group: filtered once, right after build_topology()	14
The --mode escalation rule	14
Why logs go to stderr, not stdout	14
JsonFormatter: what ends up in one log line	14
--manual: preferring man(1), falling back only when necessary	15
The Proxmox VE API client	15
Why proxmoxer	15
build_client(): the one place auth is assembled	15
PveClient: one method per section 3.5 endpoint	15
move_disk(): the one and only bytes/s -> KiB/s conversion	16

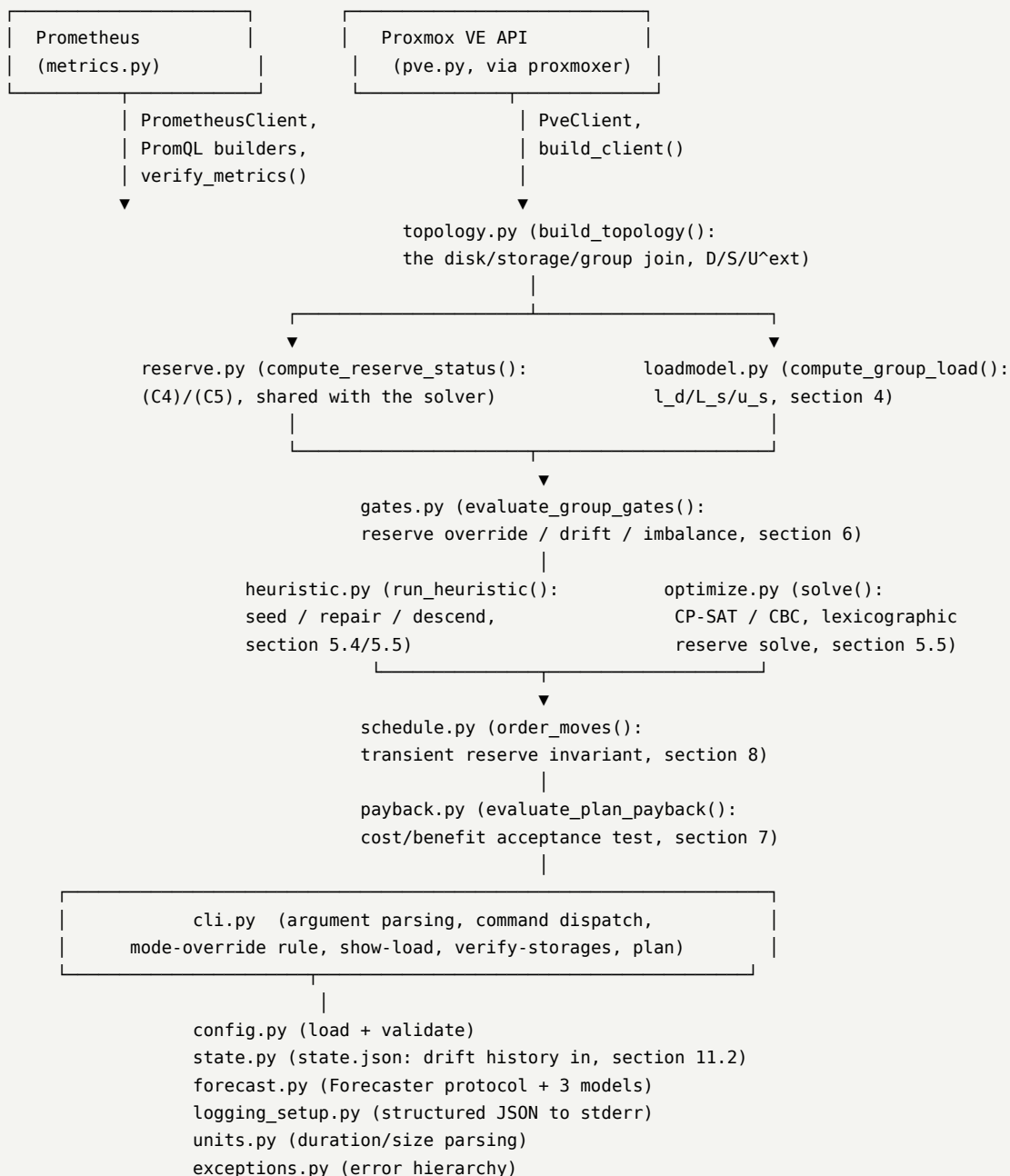
Building the disk/storage/group model	17
D means every disk this module places, pinned or not	17
One pass, in the order section 3.5 lists	17
The lock field comes from <code>vm_config()</code> , not a separate call	18
Pin priority: <code>_pin_reason()</code>	18
The per-disk cooldown pin	18
Sizes: content is authoritative, config is the fallback	18
<code>U_sext</code> : everything not referenced	18
<code>reserve.py</code> : one (C4)/(C5) evaluator, shared	19
The load model	19
One function, one group, already-built input	19
Six queries, not six-times-groups queries	19
Coverage rejection excludes a disk from the group total, not just from its own <code>l_d</code>	19
<code>tpmstate0/unusedN</code> are never rejected; <code>efidisk0</code> is	20
<code>T_g = 0</code> is “idle”, computed only from accepted disks	20
<code>L_s/u_s</code> are the <i>current</i> assignment, not a candidate one	20
What is still missing from phase 3	20
Gating: deciding whether to act at all	20
One function, three gates, in the order section 6 lists them	21
<code>last_load</code> is real now, for <code>show-load</code> and <code>plan</code>	21
<code>show-load</code> ’s gate line is diagnostic, not a real decision	21
Cooldowns are not this module’s job	21
The heuristic solver	22
<code>evaluate_assignment()</code> is the objective, kept separate from the search	22
Repair is unconditional, not weight-driven	22
Descend explores swaps, not only single moves	22
The storage cooldown excludes a destination, never a source	23
<code>objective.spread_metric</code> : two different quantities, not one rescaled	23
Proof this reproduces the plan, not just itself	23
What this pass deliberately does not do	24
The MILP backends: CP-SAT and CBC	24
Both backends solve, then hand off to <code>heuristic.evaluate_assignment()</code>	24
Lexicographic, not big-M — and why that is the only mode implemented	24
CP-SAT: section 5.5’s exact integer scaling	25
CBC: “direct transcription”, deliberately not scaled	25
<code>(C1)/(C3)/(C4)/(C5)</code> are shared code, factored once per backend	25
<code>cli.py</code> ’s dispatch: auto cascades, an explicit backend still falls back	26
Deliberately not implemented in this pass	26
Ordering the moves	26
One loop, reusing <code>heuristic.py</code> ’s objective for the ratio	26
<code>cost_m</code> is <code>z_d</code> — and that is not an approximation today	27
The transient invariant, called with a single-move set always	27
The heuristic can accept a residual violation; the scheduler cannot execute one	27
<code>final_assignment</code> is the schedule’s own truth, not the heuristic’s target	27
What this pass deliberately does not do	28

Wired into plan, not yet into apply	28
Migration cost and the payback test	28
Cost and benefit are computed from data other modules already have	28
headroom_src/headroom_dst: a plan formula this project cannot fill in	28
The reserve-override exemption	29
What this pass deliberately does not do	29
Wired into plan, not yet into apply	29

What does this page answer? Where does an invocation of `pve-storage-drs` go, which module owns which piece, and how much of `IMPLEMENTATION_PLAN.md`'s section 2 architecture actually exists right now?

The pipeline, as specified and as built

`IMPLEMENTATION_PLAN.md` section 2 describes seven stages: collect, join, gate, solve, cost, order, execute. As of this page, stages 1 (collect) through 6 (cost/order) exist, with **two** interchangeable stage-4 (solve) backends — the dependency-free heuristic and the MILP (CP-SAT/CBC) path — selected by `solver.backend`; only execute (`IMPLEMENTATION_PLAN.md` section 12 phase 7+) is not yet written. Do not take this page as a claim that the whole pipeline runs end to end — [../manual/30-safety-and-status.md](#) is the authoritative per-command status: `plan` is real and prints an ordered, transient-feasible, payback-checked dry-run plan (see [../manual/27-plan.md](#)), but nothing yet *executes* one (no `apply`). `state.json` itself exists (`state.py`) and is read by both `show-load` and `plan` for real drift history — see [15-state.md](#) and [80-gates.md](#) — but nothing writes it yet, since only `execute.py` (not yet written) has a reason to.



Module layout, as it stands

Module	Responsibility	Plan section
exceptions.py	The DrsError hierarchy every deliberate failure raises	AGENTS.md section 5
units.py	Duration/size parsing ("24h", "200MiB") and human-readable formatting	section 11
config.py	Load /etc/pve/drs.yaml, jsonschema + semantic validation, the frozen dataclass config model	section 11
config_schema.json	The jsonschema structural half of validation	section 11.1
state.py	state.json: the load vector as of the last executed balance, cooldowns, the node-local advisory flock() — reading degrades, writing raises	section 11.2
forecast.py	The Forecaster protocol, required_range_seconds, and quantile/seasonal_naive/holt_winters	section 10
logging_setup.py	Structured JSON logging to stderr	section 2.1 (amended, see 40-cli-and-logging.md)
metrics.py	PrometheusClient, PromQL construction, verify_metrics(), compute_disk_coverage()	sections 3.1-3.4
pve.py	PveClient (built on proxmoxer), build_client()	section 3.5
topology.py	build_topology(): the disk/storage/group join, D, S, U ^{s e x t} , (C2) pins	sections 3.5-3.7, 5.1, 5.3 (C2)
reserve.py	compute_reserve_status(): (C4)/(C5), shared by show-load today and the solver later	section 5.3 (C4)/(C5)
loadmodel.py	compute_group_load(): the raw-series-to- <i>l_d</i> blend, min_coverage rejection, current L _s /u _s	section 4
gates.py	evaluate_group_gates(): reserve override, drift, imbalance — the act/no-act verdict, with reasoning	section 6
heuristic.py	run_heuristic(): seed/repair/descend, and evaluate_assignment(), the section 5.4 objective shared with the MILP path too	sections 5.4/5.5
optimize.py	solve(): CP-SAT/CBC, section 5.3's constraints, the lexicographic two-stage reserve solve	section 5.5
schedule.py	order_moves(): transient-feasible ordering of a target assignment's moves, deadlock reporting	section 8
payback.py	evaluate_plan_payback(): the cost/benefit acceptance test, with a reserve-override exemption mirroring gates.py's	section 7
cli.py	Argument parsing, command dispatch, --manual, the mode-override rule, show-load, verify-storages, plan	section 11.3

Not yet written: execute.py (phase 7+) and auto mode (phase 8) – optimize.py (phase 6) is done: plan

picks CP-SAT, CBC or the heuristic per solver.backend, see [91-optimize.md](#). Cooldowns are now read: topology.py's (C2) per-disk pin and heuristic.py's per-storage target exclusion both consume state.py's cooldown data (see [15-state.md](#), [60-topology.md](#) and [90-heuristic.md](#)) — only *writing* a cooldown still waits on execute.py, and the MILP path does not enforce the storage cooldown at all yet (see [91-optimize.md](#)). Within modules that do exist: heuristic step 4 “polish” and (C2) format-compatibility eligibility in heuristic.py (see [90-heuristic.md](#)), concurrent scheduling, priority-2 ordering and staging in schedule.py (see [95-schedule.md](#)), and the payback re-solve-and-retry loop plus the section 7.3 saturation check in payback.py (see [96-payback.md](#)).

Why config.py depends on forecast.py

config.py's semantic validation (`_check_forecast_window`) must reject a window.lookback too short for the configured forecast.model at startup (section 11.1) — that check needs forecast.required_range_seconds(). Since forecast.py in turn needs config.ForecastConfig for its type signatures, the import is deferred to inside the validation function rather than at module level, breaking what would otherwise be an import cycle. See the comment at the top of config.py's `_check_forecast_window` if you touch this.

Where to read next

- [10-configuration.md](#) — how a YAML file becomes a validated Config, and where every default actually lives.
- [15-state.md](#) — state.json's dataclasses, why reading it degrades but writing raises, the real flock() lock and the rename-vs -in-place bug it takes to get that wrong, what show-load/plan actually get from it today, and how its cooldown data reaches topology.py/heuristic.py.
- [20-forecasting.md](#) — the forecaster protocol and its three implementations.
- [30-metrics.md](#) — the Prometheus client and the six verify-metrics checks.
- [40-cli-and-logging.md](#) — command dispatch, the --mode escalation rule, and why logs go to stderr.
- [50-pve-api.md](#) — the PVE API client, why it is built on proxmoxer, and two things verified against a real cluster.
- [60-topology.md](#) — the disk/storage/group join, the section 5.1.1 foreign-volume accounting, the shared (C4)/(C5) evaluator, and the per-disk cooldown pin.
- [70-loadmodel.md](#) — the section 4 raw-series-to-l_d blend, min_coverage rejection, and why tpmstate0/unusedN are exempt from it but efidisk0 is not.
- [80-gates.md](#) — the section 6 act/no-act verdict, how state.json's drift history now reaches it, and why the per-disk/ per-storage cooldowns live in topology.py/heuristic.py instead.
- [90-heuristic.md](#) — the section 5.4/5.5 objective and the seed/repair/descend search, cross-checked against section 14's exact objective totals, the storage-cooldown destination filter and its repair-side exemption, and what “polish” and format eligibility still owe.
- [91-optimize.md](#) — the CP-SAT and CBC MILP backends, section 5.5's exact integer coefficient scaling (and the γ -term trap it exists to avoid), why the reserve is solved in two lexicographic stages rather than one big-M objective, and cli.py's solver.backend dispatch (auto's cascade, and why an explicitly forced backend still falls back to the heuristic rather than failing).
- [95-schedule.md](#) — ordering a target assignment's moves under the section 8 transient invariant, why cost_m = z_d is exact today (not an approximation), and why the heuristic accepting a residual violation can still mean the scheduler reports a deadlock.
- [96-payback.md](#) — the section 7 cost/benefit test, the headroom_src/headroom_dst gap in the plan's own cost formula, and why a reserve-fixing plan always passes the economic test.

How a YAML file becomes a validated Config

What does this page answer? What exactly happens between pve-storage-drs -c foo.yaml plan and a type-checked Config object, and where does each of the section 11.1 validation rules actually live?

Describes `proxmox_storage_drs/config.py` and `config_schema.json`.

Resolution order

`resolve_config_path()` implements `IMPLEMENTATION_PLAN.md` section 11's three-source order: `--config` (`cli_path`), then `$PVE_STORAGE_DRS_CONFIG`, then `config.DEFAULT_CONFIG_PATH` (`/etc/pve/drs.yaml`). It also returns `was_explicit`, which `load_config()` uses to decide the error message on a read failure: an explicitly named path that cannot be read is always fatal with no fallback, per section 11, but the *message* additionally points at `config/drs.example.yaml` only when the failing path was the unnamed default — there is nothing useful to suggest when the operator named the path themselves.

The two-stage validation

1. `_validate_schema()` loads `config_schema.json` via `importlib.resources` (it ships as package data — see `pyproject.toml`'s `[tool.setuptools.package-data]`) and runs it through `jsonschema`. Every object in the schema sets `additionalProperties: false`: a typo'd key is caught here as a validation error, not silently ignored, which is what keeps section 15.1's knob-to-formula table trustworthy (a key with no formula would otherwise go unnoticed).
2. `_build_config()` walks the now-schema-valid raw dict into the frozen dataclasses defined earlier in the module, applying every default inline via `.get(key, default)`. These defaults are also what `config/drs.example.yaml` documents and what the manual's [../manual/10-configuration.md](#) describes — there is deliberately no second copy of a default anywhere else in the codebase.
3. `_validate_semantics()` runs the section 11.1 rules `jsonschema` cannot express — cross-field comparisons, or rules needing a value derived from two independently-optional settings. Each rule is a small `_check_*` function appending to a shared `errors/warnings` list (kept separate functions rather than one large one, partly for `flake8`'s max-complexity and partly because each is independently testable). `errors` become one `ConfigError` naming every problem found, not just the first; `warnings` are returned to the caller (`ResolvedConfig.warnings`) for `cli.py` to log — section 11.1's `saturation_load` check is the one warning-only rule today.

Every duration or size field is parsed **once**, at this point, via `units.parse_duration_seconds/parse_size_bytes`, and stored on the dataclass already in its canonical unit (seconds, bytes) — nothing downstream re-parses a string.

Secrets from the environment

`_build_config()` fills `proxmox.auth.password/token_secret` from `PVE_PASSWORD/PVE_TOKEN_SECRET` **only when the file's own value is null** — a secret actually written in the file is used as-is, never silently overridden by an environment variable a deploy script forgot to unset. This one-directional fallback is what section 11's "keep secrets out of `/etc/pve`" guidance is for: the file can omit the field entirely and rely on the environment, or set it explicitly and take responsibility for the file's own permissions.

What is deliberately *not* checked here

Two section 11.1 rules need live cluster data this module does not have and are therefore **not** part of `config.py`:

- storage ids actually existing in the cluster;
- `execution.source_release.timeout` / `gates.cooldown_per_storage` against a storage's real `saferemove_throughput`.

Both belong to `pve.py/topology.py` (which do the live cluster fetch) and `pve-storage-drs verify-storages` — see [../manual/30-safety-and-status.md](#) for current status.

ResolvedConfig: what the caller actually gets

`load_config()` returns a `ResolvedConfig`, not a bare `Config`: the resolved path and the file's sha256 travel with it, because `IMPLEMENTATION_PLAN.md` section 11 requires every run to log both — a plan must always be traceable to the exact file that produced it, even though the file is never re-read after startup.

Persistent state: `state.json`

What does this page answer? What does `state.py` actually store, how does reading it degrade instead of failing, why is the advisory lock a real `flock()` rather than a JSON field, and what does today's wiring into `show-load/plan` actually change? Describes `proxmox_storage_drs/state.py`.

One dataclass tree, matching section 11.2's JSON exactly

State mirrors the plan's own example document field for field: `lock (LockInfo | None)`, `last_balance` (at plus a "`<group>:<vmid>:<device>`" -> `l_d load_vector`), `cooldowns` (`disk/storage`, each a key -> timestamp mapping), `inflight_upids`, `staged_disks`. `disk_state_key()/storage_state_key()` build the two key shapes section 11.2 specifies — group-prefixed specifically so a vmid reused after a VM is destroyed and recreated in a *different* group cannot collide with its old entry. `empty_state()` is what a fresh install, or a lost file, degrades to: no lock, no balance, no cooldowns, nothing in flight.

Reading degrades; writing does not

`load_state()` never raises. A missing file is the ordinary first-run case (silent); a present-but-corrupt, empty, or wrong-schema_version one is still just `empty_state()`, but logged at `WARNING` — section 11.2's own words are “losing this file is safe but not free: cooldowns and drift history reset”, not “state.json existing and being readable is required to run”. This mirrors `show-load`'s choice to degrade a group's load to “unavailable” on a Prometheus outage rather than fail the whole command — neither Prometheus nor `state.json` is an *explicitly requested* input the way `-c/--config PATH` is, and only the latter is a hard failure on error (`.agents/domain-invariants.md` section 9).

`save_state_atomic()`, by contrast, raises `StateError` on any failure — writing (or locking) is something a caller actively asked this module to do, not a read with a documented, safe fallback. It writes to a temp file in the same directory, `fsyncs`, then `os.replace()`s over the real path, so a concurrent `load_state()` can never observe a half-written file — which is exactly what lets every read in this codebase skip locking entirely (next section).

Locking: `flock()` is the mechanism, the JSON `lock` field is a label

`acquire_lock()` takes a real, non-blocking `fcntl.flock(LOCK_EX | LOCK_NB)` on the state file itself. That is the actual, kernel-enforced mutual exclusion section 11.2 asks for, and it is correct by construction for two instances on the *same host* — `.agents/domain-invariants.md` section 9 is explicit that “the `fcntl` lock cannot see another node”; cross-node coordination is the separate in-flight-UPID scan (section 13), not this module's job.

The `lock: {pid, host, acquired_at}` field recorded *inside* the JSON is **descriptive metadata for a human reading the file, not the exclusion mechanism** — by the time this module's code overwrites it, the OS has already granted exclusive ownership, so whatever a previous run last wrote there is necessarily stale and is unconditionally replaced, never inspected to decide whether to proceed. This is also how section 11.2's “if `lock.pid` is not alive on `lock.host`, reclaim it” is satisfied, without a separate liveness check to get subtly wrong: a process that died released its `flock()` the moment its file descriptor closed (any exit, including a crash), so the *next* `acquire_lock()` call simply succeeds. `acquire_lock()` returns `None` — not an exception — when another live instance already holds it, matching section 11.2's “a live PID means another instance is running: exit 0 quietly”; deciding what “quietly” means, and which exit code, is left to the caller.

A real bug this design surfaced during testing, worth remembering if you touch this code: the first implementation had `acquire_lock()` write its lock metadata via `save_state_atomic()` — the same temp-file-plus-rename used everywhere else. That silently breaks the lock: a rename replaces the directory entry with a *new* inode that was never `flock()`'d, so a second `acquire_lock()` opens that new inode fresh and locks it with no conflict at all — `test_a_second_acquire_while_the_first_is_held_returns_none` caught this immediately. The fix, `_write_state_to_locked_fd()`, writes **in place** into the already-open, already-locked file descriptor (`lseek` to 0, `write`, `ftruncate` to the new length, `fsync`) — never a rename — for as long as the lock is held.

What today's wiring into `cli.py` actually changes

`show-load` and `plan` both call `_last_loads_by_group()` once, right after reading configuration, and use the result for both `loadmodel.compute_group_load()`'s `last_known_loads` and `gates.evaluate_group_gates()`'s `last_load` — one state read per run, one function computing the per-group slice, not two independent implementations (AGENTS.md section 5). Neither command takes the advisory lock: both are read-only reports that never execute a migration, and taking the exclusive lock for one would make an in-progress apply block `plan/show-load` for no safety reason the `plan` text supports.

The practical effect: a `state.json` with a real `last_balance` now makes the drift gate genuinely suppress an ACT verdict the imbalance alone would otherwise trigger — see `test_show_load_gate_reflects_real_drift_history_from_state_json` and `test_plan_gate_also_reflects_real_drift_history_from_state_json` in `tests/unit/test_cli.py` for the exact before/after. Without a `state.json` on disk (the common case until `execute.py` exists — see below), behaviour is unchanged from before this module existed: `last_load=None`, drift gate skipped, “reserve override, else imbalance”.

Cooldowns: read by `topology.py` and `heuristic.py`, not by this module

`state.py` only stores and queries cooldown timestamps (`active_disk_cooldowns()/active_storage_cooldowns()`); it does not decide what a cooldown *means* to the solver. `build_topology()` (via a new `state/now` parameter, both defaulting to “no cooldowns”/“real clock”) computes each group's active disk cooldowns once and feeds them into `topology._pin_reason()` — section 5.3 (C2)'s “d is within its per-disk cooldown -> also pin to current”, reported as `cooldown: moved recently, <time> left on gates.cooldown_per_disk`, in the plan's own priority order (after a snapshot pin, before a lock pin). `cli.py` separately computes each group's active *storage* cooldowns and passes them to `heuristic.run_heuristic()`, which excludes them as a move/swap **destination** in `_descend()` only — see `docs/internals/60-topology.md` and `docs/internals/90-heuristic.md` for the two halves in detail, including the deliberate asymmetry: a live (C4)/(C5) repair move ignores the storage cooldown entirely (section 13's reserve-override principle), while nothing yet exempts a disk-cooldown pin from blocking a repair the same way — a known, documented limitation, not an oversight.

Like `last_balance`, this is **read-only wiring**: nothing calls `with_recorded_cooldown()` yet, so every cooldown query returns empty until `execute.py` records one.

Deliberately not implemented in this pass

- **Nothing calls `acquire_lock()` or `with_recorded_balance()/with_recorded_cooldown()` yet.** All exist and are fully tested, but only `execute.py` (phase 7, not yet written) has a reason to take the lock or record anything — section 11.2 itself says `last_balance` is “updated only after a run that executed at least one migration”, and neither `show-load` nor `plan` ever does.
- **A disk-cooldown pin is not exempted for an active reserve violation the way the storage cooldown is.** `topology.py` decides a disk's pin before the group's `reserve.ReserveStatus` is even computable (it needs every disk in the group, which is still being built), so there is no cheap way today to ask “would un-pinning this disk help fix a live violation?” at the point the pin decision is

made. The practical effect is bounded and safe, never silently wrong: `heuristic._repair()` already reports an unresolvable residual violation via `reserve_penalty_term` whenever no movable disk can fix it (the same path `test_repair_does_not_oscillate_when_no_target_can_fully_absorb_the_violation` exercises) — a disk-cooldown pin can only ever add to the set of cases that hit that path, never bypass it. Revisiting this needs restructuring `topology.py` into two passes (build every disk first, compute reserve status per storage, then finalize cooldown pins), which is a real, separately-scoped change.

- **inflight_upids/staged_disks** exist in the dataclass and round-trip correctly, but nothing writes or reads them yet — they belong to `execute.py` (crash recovery, section 13) and `schedule.py`’s staging (section 8.3 option 1, also not implemented), respectively.

Forecasting: the protocol and its three implementations

What does this page answer? How does `forecast.py` decide how much history a model needs, and what does each of `quantile/seasonal_naive/holt_winters` actually compute? Describes `proxmox_storage_drs/forecast.py`.

The one rule, in one place

`IMPLEMENTATION_PLAN.md` section 10.1’s table says how much history each model needs, independent of `window.lookback` (the *decision* window). That rule has exactly one implementation: three small pure functions, `_quantile_required_range_seconds`, `_seasonal_naive_required_range_seconds` and `_holt_winters_required_range_seconds`, each called from **two** places — the free function `required_range_seconds()` that `config.py`’s startup validation uses (before any Forecaster is constructed), and the matching `Forecaster.required_range()` method on the concrete class. Duplicating the arithmetic between those two call sites, rather than sharing the pure function, is exactly the kind of “second copy of a rule” `AGENTS.md` section 5 forbids — a change to the Holt-Winters requirement made in only one of them would silently desynchronize config validation from what the forecaster itself believes it needs.

Why the upper bound, never the point estimate

Every `Forecaster.predict()` returns a `Forecast(point_estimate, upper_bound)`. Nothing in this module enforces that callers use `upper_bound` — that discipline belongs to the caller (the optimizer and the section 7.3 saturation guard, neither written yet). The asymmetry is stated in the module docstring and repeated here because it is easy to get backwards under time pressure: overestimating costs a slightly worse balance, underestimating risks scheduling a mirror onto a storage that is about to saturate.

QuantileForecaster

The default. `predict()` takes the `window.quantile/window.upper_quantile` percentiles of the raw series with no fitting at all. `_quantile()` implements linear-interpolation percentile matching `numpy.percentile`’s default method, by hand — deliberately not using `numpy`, since the tool must stay usable with only `requests + ruamel.yaml + jsonschema` installed (`IMPLEMENTATION_PLAN.md` section 2.1). `horizon` is accepted (to satisfy the Forecaster protocol) and immediately discarded: the quantile model has no notion of a horizon-dependent forecast.

SeasonalNaiveForecaster

Groups historical samples by hour-of-day (`int((ts // 3600) % 24)`) and takes the median as the point estimate, the configured upper quantile across the same bucket as the bound. `now_epoch_seconds` selects which hour-of-day bucket the *current* moment falls into; `predict()` ignores everything outside that bucket. A bucket with no samples returns `Forecast(0.0, 0.0)` rather than raising, matching `IMPLEMENTATION_PLAN.md` section 4’s rule for an idle group’s raw quantities: no data is zero, not an error, at this layer (the caller decides whether zero is trustworthy).

HoltWintersForecaster

`statsmodels` is an optional dependency (Suggests, not Depends — see `debian/control`), so it is imported **only inside `predict()`**, never at module level; `debian/tests/import-all` is what would catch a regression here. Two situations fall back to `QuantileForecaster`, both logged at warning level rather than failing silently:

1. fewer than $2 * \text{seasonal_periods}$ samples in the series (section 10.2's own rule — a fit on too little data is worse than no fit);
2. `statsmodels` is not importable at all.

The upper bound is `point_estimate + residual_z * stdev(in-sample residuals)` — not derived from the model's own confidence interval, which `statsmodels's ExponentialSmoothing.fit()` does not expose directly for every configuration. `residual_z` (`forecast.holt_winters.residual_z`, default 2.0) is what an operator tunes if this bound turns out too tight or too loose in practice; there is no plan-mandated backtesting gate on it yet (`IMPLEMENTATION_PLAN.md` section 10.2's backtest-validation requirement is not implemented — tracked for the dedicated forecasting phase, section 12 phase 9).

build_forecaster()

The one factory function that turns a `ForecastConfig` plus the ambient `window/metrics` values into a live `Forecaster` instance. Nothing else in the codebase should construct a `QuantileForecaster/SeasonalNaiveForecaster/HoltWintersForecaster` directly once a caller exists that needs one from config — route it through here so the model-name-to-class mapping stays in one place.

The Prometheus client and verify-metrics

What does this page answer? How does `metrics.py` talk to Prometheus without ever touching the network in a test, and what does each of the six `verify-metrics` checks actually query? Describes `proxmox_storage_drs/metrics.py`.

_SessionLike / _ResponseLike: why not requests.Session directly

`PrometheusClient.__init__` accepts `session: _SessionLike | None`, a Protocol defined in this module with only the one `get(...)` method the client actually calls — not `requests.Session` itself. `.agents/testing.md` forbids any test talking to a real Prometheus; a structural protocol lets a test hand in a small dataclass with a matching `get()` instead of mocking or subclassing the real (network-capable) session. `PrometheusClient()` with no session argument still constructs a real `requests.Session()`, which satisfies the protocol structurally — nothing special is needed to make the real thing fit.

_get(): one request path, three endpoint shapes

Every one of `instant_query`, `range_query` and `label_values` goes through the private `_get()`, which does the HTTP call, checks the status code, parses JSON, checks Prometheus's own status: "success" field, and returns `payload["data"]`. That field's *shape* genuinely differs by endpoint — a dict with a result list for `query/query_range`, a bare list for `label/<name>/values` — which is why `_get()` is typed `Any` rather than picking one shape; each public method knows which shape it asked for and narrows accordingly. Every failure mode (a transport error, a non-200 status, unparseable JSON, an unsuccessful Prometheus response) raises `MetricsError` from exactly this one place, so a caller catching `MetricsError` around any of the three public methods sees every failure mode uniformly.

PromQL construction

`build_rate_promql()` and `build_quantile_over_time_promql()` are pure string-building functions, deliberately factored out of any query-issuing code so they are unit-testable without a fake session at all —

see `IMPLEMENTATION_PLAN.md` section 3.4 for the exact expressions they implement. `raw_metric_name()` is the one place that maps a `RAW_METRIC_FIELDS` entry ("`read_ops`", ...) to the configured metric name on a `MetricsConfig`, via `getattr` — this is what lets `verify_metrics()` iterate “every configured raw metric” without hand-listing the six field names a second time anywhere in this module.

`verify_metrics()`: six checks, six functions

Each `_check_*` function implements exactly one `IMPLEMENTATION_PLAN.md` section 3.3 numbered check and returns `Findings` (plus, where relevant, the data the report carries forward):

Function	Section 3.3 step	Returns
<code>_check_metric_names_exist</code>	1	findings only
<code>_check_sample_series</code>	2 + 3 (labels present)	findings, { <code>metric_name</code> : <code>sample_labels</code> }
<code>_check_device_label_collision</code>	4	one <code>Finding</code> or <code>None</code> (pure, no I/O)
<code>_check_coverage</code>	5	findings, { <code>DiskKey</code> : <code>coverage_fraction</code> }
<code>_check_observed_spacing</code>	6	findings, <code>observed_spacing_seconds</code>

`_check_coverage` and `_check_observed_spacing` both use `metrics.read_ops` as the one representative metric rather than probing all six: a coverage or spacing gap is a property of the underlying Telegraf scrape, not of which of the six raw quantities is read, so probing all six would be six times the Prometheus load for no additional information.

`VerifyMetricsReport.ok` is `True` iff no `Finding` has `level == "error"` — `cli.py`’s `verify-metrics` handler uses exactly this property to decide the process exit code, so there is one definition of “the verification passed.”

What `cli.py` does with the report

`_handle_verify_metrics` in `cli.py` constructs a `PrometheusClient` from `resolved.config.prometheus`, calls `verify_metrics()`, and renders either the human form (`[level]` message lines) or, under `--json`, a JSON-serializable dict — `DiskKey` keys are turned into "`vmid:device`" strings there, since JSON object keys must be strings and this module keeps no opinion about output formatting.

Command dispatch, the mode-override rule, and why logs go to `stderr`

What does this page answer? How does `cli.py` turn `argv` into a dispatched command, what exactly does overriding `--mode log`, and why does structured logging go to `stderr` instead of the `stdout` the plan originally specified? Describes `proxmox_storage_drs/cli.py` and `proxmox_storage_drs/logging_setup.py`.

One parser, global options first

`build_parser()` defines every global option (`-c/--config`, `--group`, `--mode`, `--json`, `-v/--quiet`, `--version`, `--manual`) on the **top-level** `ArgumentParser`, not on the subparsers, which is what makes `argparse` itself enforce `IMPLEMENTATION_PLAN.md` section 11.3’s “global options before the subcommand” for free — there is no hand-written ordering check anywhere. `--help`’s defaults come from the real default= values passed to `add_argument`, combined with `argparse.ArgumentDefaultsHelpFormatter`; there is deliberately no second, hand-maintained list of options and defaults anywhere in this module, the manpage, or the manual (`AGENTS.md` section 8.5).

`main()` handles `--version` and `--manual/help` **before** requiring or loading any configuration — printing the version or the manual page must never depend on `/etc/pve/drs.yaml` existing. Only once a real subcommand is present does it call `config.load_config()`.

Command handlers: a dict, not a chain of ifs

`_COMMAND_HANDLERS`: `dict[str, CommandHandler]` maps each subcommand name to a function of `(ResolvedConfig, argparse.Namespace, effective_mode) -> int`. Every subcommand not yet implemented (apply, explain — see [../manual/30-safety-and-status.md](#) for which ones that currently is) gets a handler from `_make_not_yet_implemented_handler(name)`, a closure factory rather than a lambda with a default-argument trick, because mypy's strict mode cannot infer a bare lambda's parameter types cleanly here — see the comment at that function if you are tempted to simplify it back to a one-liner. `_COMMAND_HANDLERS["verify-metrics"]` is overwritten with the real `_handle_verify_metrics` once `metrics.py` exists to back it. Adding a new implemented command is exactly this: write the handler, assign it into the dict, done — `main()`'s dispatch does not change.

--group: filtered once, right after build_topology()

`_filter_groups(topology, args.group)` is the one place `--group` is actually read. `show-load`, `verify-storages` and `plan` each call it immediately after `build_topology()`, replacing that `Topology`'s groups tuple with the (still topology-order) subset named — every render function downstream just iterates `topology.groups` as before and needs no `--group` awareness of its own. `verify-metrics` does not call it: that command validates configured metric/label names against Prometheus directly and never iterates groups at all, so there is nothing for `--group` to restrict there. A name that matches no configured group raises `DrsError` rather than silently producing an empty report — the same “an explicitly named thing that doesn't resolve is a hard failure” rule `config.load_config()` already applies to `-c/--config PATH` (REVIEW.md R-03: `--group` was previously parsed and documented, but no handler ever read `args.group` at all).

The --mode escalation rule

`apply_mode_override(configured_mode, override)` is the entire implementation of section 11.3's rule: moving *down* the ordering `dry-run < confirm < auto` (toward more caution) logs at `INFO`; moving *up* it (toward less caution — removing a barrier the operator's own config set) logs at `WARNING`, naming both values. This is deliberately a small pure-ish function (its only side effect is the log call) rather than inlined into `main()`, so it is unit-testable against a caplog fixture without constructing a full CLI invocation.

Why logs go to stderr, not stdout

`IMPLEMENTATION_PLAN.md` section 2.1 originally specified `stdout` for structured JSON logs. This was amended in the same commit that added `logging_setup.py`: `--json` (section 9.5) emits the plan report on **stdout**, and interleaving log lines with that report on the same stream would make it unparseable by anything downstream. Logging instead goes to **stderr**; a systemd service unit captures both streams into the same journal regardless, so nothing about “captured by journald” is lost. This is the one place in the codebase where the implementation and `IMPLEMENTATION_PLAN.md` disagree with the *original* plan text on purpose — `AGENTS.md` section 7 rule 2 is why the plan text itself was corrected in the same commit rather than left to silently diverge from the code.

`configure_logging()` is called exactly once, from `main()`, after the `--version/--manual` early exits (which must not touch logging configuration at all) and before any config-dependent code runs, so that a config-loading failure is itself logged consistently with everything after it.

JsonFormatter: what ends up in one log line

Every field on a `logging.LogRecord` that is *not* one of the standard attributes (computed once, at import time, into `_STANDARD_RECORD_ATTRS`) is folded into the JSON payload — this is what lets any module call `logger.info(..., extra={"event": "gate_decision", "threshold": 0.2})` and have `event/threshold` appear as top-level JSON keys with no formatter changes required. `sort_keys=True` keeps output byte-stable for anything that diffs log lines in a test or a log-aggregation pipeline.

--manual: preferring man(1), falling back only when necessary

`show_manual()` tries `man pve-storage-drs` first, via `subprocess.run` rather than `os.execvp`: `man(1)` itself detects a non-tty stdout and disables its pager automatically, which is what satisfies “never answer with a URL alone” (AGENTS.md section 8.5) whether the invocation is interactive or piped — there is no need for this module to duplicate that tty detection. `_fallback_manual_text()` is reached only when `man(1)` itself is missing or has no entry (an uninstalled source checkout, or a minimal system with no `man-db`): it walks up from `cli.py`’s own file location looking for `man/pve-storage-drs.1.md` in a source tree, and only if that also fails prints a short message naming the real installed-package path and the in-tree file — a local, actionable path, never a bare URL.

The Proxmox VE API client

What does this page answer? Why is `pve.py` built on `proxmoxer` instead of a hand-rolled ticket/CSRF client, and what does each of its methods actually call? Describes `proxmox_storage_drs/pve.py`.

Why proxmoxer

IMPLEMENTATION_PLAN.md section 3.5 explains the decision in full; the short version is `proxmoxer`’s **backend abstraction**. The same `ProxmoxAPI` attribute-chaining interface (`api.nodes(node).qemu(vmid).config.get()`) is available over plain HTTPS (what `build_client()` uses today) or over SSH, either shelling out to the system’s own `ssh+pvesh` (`openssh`) or with an in-process client (`ssh_paramiko`). A deployment that cannot open the API port to the management host becomes a different keyword argument in `build_client()`, with no change anywhere else — not to `PveClient`’s methods, and not to any of their callers.

build_client(): the one place auth is assembled

`config.py`’s `ProxmoxConfig` keeps `verify_ssl` (bool) and `ca_file` (path-or-None) as two separate knobs, because that is how an operator thinks about them; requests (and so `proxmoxer`’s https backend) wants one value, a bool or a CA-bundle path, passed as `verify`. `build_client()` is the one place that reconciles the two. It is also the one place `proxmox.auth.token_id`’s `user@realm!tokenname` format is split into `proxmoxer`’s separate `user/token_name` keyword arguments — never duplicate that parsing anywhere else.

Every failure `ProxmoxAPI(...)` itself can raise (`proxmoxer.AuthenticationError`, or a `requests.RequestException` if the host is simply unreachable) is caught here and re-raised as `PveApiError`, so every caller of `build_client()` and every `PveClient` method has exactly one exception type to catch.

PveClient: one method per section 3.5 endpoint

Every method is a single API call, wrapped through the private `_call()` helper, which is the one place `proxmoxer.ResourceException` (HTTP-level API errors) and `requests.RequestException` (transport failures `proxmoxer`’s https backend does not wrap itself) both become `PveApiError`. No method caches anything, and the per-run topology cache belongs to `topology.py` (section 3.5), which is also where the bounded thread pool for the O(VMs) config fetch lives (60-topology.md, REVIEW.md P-02).

`_call()` does retry exactly one failure mode: `proxmoxer.AuthenticationError` gets one reauthenticate-and-retry cycle (REVIEW.md P-01) before becoming a `PveApiError`, via a reauthenticate callback `build_client()` wires in — a full fresh login through `_build_api()` again, not a reuse of the rejected ticket. This is not a general retry policy (a `ResourceException` or a transport failure still fails immediately, on the first attempt); it exists specifically because a ticket can expire for reasons with nothing to do with the call that hits it — a long apply --confirm wait, a suspended process, a clock jump — and `proxmoxer`’s own lazy per-request renewal only notices the age of its own clock, not whether the server-side ticket actually outlived

a gap that long. A test double built with a bare `PveClient(fake_api)` (no `reauthenticate=`) gets none of this — it behaves exactly as it did before P-01, which is what every existing fake in `tests/unit/fakes.py` relies on.

`build_client()` also applies `proxmox.ticket_refresh_seconds` to the constructed session, best-effort: proxmoxer 2.x has no constructor argument for a password/ticket auth's refresh interval (it is a hard-coded `renew_age = 3600` class attribute on `ProxmoxHTTPAuth`, checked lazily on whichever request happens to run after it elapses), so `_apply_ticket_refresh_seconds()` reaches into `api._backend.auth` — undocumented proxmoxer internals, not its public interface — to override that instance's `renew_age` after login. Deliberately narrow: it only ever *tightens* proxmoxer's own schedule, never replaces its refresh logic, and if a future proxmoxer version reshapes this (or the backend is not `https`, or the auth is API-token, which has no ticket at all), the `getattr/hasattr` guards make it a silent no-op — the P-01 `reauthenticate-and-retry` above still catches the case this can't reach.

`storage_definitions()` deliberately uses `GET /storage` (the list form), never the singular `GET /storage/{id}`, even though only one storage's config is needed in some callers. This was verified empirically against a real PVE 9.2.11 cluster during development ([[dev-cluster-access]] in the maintainer's notes, not reproduced here): an API token granted only `Datastore.Audit` can list every storage's full config — including `saferemove/saferemove_throughput` — via `GET /storage`, but the identical data via `GET /storage/{id}` for the same storage returns 403 Forbidden (`Datastore.Allocate`). The single-item route apparently backs an edit-UI flow gated by the ability to change the config, not merely read it. Since the list form returns the same per-storage object for every storage in one call, there is no reason to ever call the singular form, and using it would quietly force a Storage DRS token to hold more privilege than the tool actually needs. IMPLEMENTATION_PLAN.md section 3.5 was corrected to match once this was found — see that section's note.

`storage_content()` returned an empty list — not an error — on every storage, node and content-type filter tried, until the token also held `Datastore.Allocate` on that storage. This was genuinely surprising and worth calling out precisely because it is a *silent* false negative: `Datastore.Audit` alone gives a clean 200 with `{"data": []}`, which looks exactly like “this storage legitimately has no tracked content” rather than “the caller cannot see it.” It was not guessed at — see IMPLEMENTATION_PLAN.md section 3.5's note, which records the exact before/after (0 entries with `Audit` only, 37 and 1 entries respectively for two real storages once `Datastore.Allocate` was added) on the same live cluster. **The practical consequence: `pve.py`'s own “keep the token to Audit-only” goal for `storage_definitions()` does not extend to `storage_content()` — a Storage DRS token genuinely needs `Datastore.Allocate` (bundled with `Datastore.Audit` in one role) on every storage it manages, or `topology.py` will silently compute U^{next} (section 5.1.1) as if every storage were empty of untracked volumes, with no error to notice by.** docs/manual/00-installation.md states the exact minimum ACL grant an operator needs; this is not a place to economize on privilege out of a preference that turned out not to match reality.

Two more things about Proxmox's ACL model this surfaced, both now reflected in the manual's token-setup instructions: **an API token's effective permission is the intersection of its owning user's permissions and whatever is granted to the token itself** when privilege separation is on (the default) — granting a role to only one of the two has no effect, both need it; and the grant that was confirmed to work was made explicitly at each `/storage/{id}` rather than only at the parent `/storage` path, so that is what the manual instructs, rather than relying on propagation that was not cleanly isolated as working or not.

move_disk(): the one and only bytes/s -> KiB/s conversion

Section 9.2 is explicit that `bwlimit`'s bytes/s-to-KiB/s conversion happens “at the call site and nowhere else.” `PveClient.move_disk()` is that call site: every caller in this codebase, present and future, passes `bwlimit_bytes_per_sec` in the same unit `migration.bwlimit_bytes_per_sec` already uses, and the `divi-`

sion by 1024 (rounded, since the API wants an integer) happens exactly once, inside this method. `format` defaults to `None` and is only ever forwarded when a caller explicitly passes one — `move_disk` without `format=` preserves the source format, which is what section 3.5 requires by default.

Building the disk/storage/group model

What does this page answer? How does `topology.py` turn `pve.py`'s raw API responses into the per-group D/S sets the rest of the engine needs, and where does each `IMPLEMENTATION_PLAN.md` section 5.3 (C2) pin condition actually get decided? Describes `proxmox_storage_drs/topology.py` and `proxmox_storage_drs/reserve.py`.

D means every disk this module places, pinned or not

This is stated at the top of `topology.py`'s own docstring because it is the single easiest thing to get backwards: D (section 5.1) is *every* disk this module puts into a group's `Group.disks`, whether or not (C2) pins its placement. A disk this module never sees at all — a stopped VM excluded by `exclude.running_only`, or a disk whose current storage is not in any configured group — is what “foreign” (`Useext`, section 5.1.1) means. Config-excluded disks (`exclude.vmid`s/`exclude.disks`/`tags`) are *not* foreign: (C2) pins them into D specifically so their bytes still count in (C4)/(C5) and their fragmentation toward κ (section 3.6, “Pinned disks are modelled, not ignored”). An earlier draft of section 5.1.1 listed config-excluded disks as foreign, which directly contradicted (C2) — that self-contradiction was found and fixed in the same commit that first implemented this join; see that section's note if you need the history.

One pass, in the order section 3.5 lists

`build_topology()` fetches everything it needs exactly once, in five stages (`_fetch_cluster_data`, then a loop over `client.vm_resources()` calling `_collect_vm_disks` per VM, then `_build_storages`):

1. `storage_definitions()` (the list form — see 50-pve-api.md), validated against the configured groups (`_validate_group_storages`): a storage that does not exist, or that lacks images in its content types, is a fatal `TopologyError` (section 11.1: “storage ids exist in the cluster”). A storage that exists but is not marked shared is a warning, not a fatal error — plausible, if unusual, for a single-node group.
2. `storage_resources()` once, to pick one active node per storage (`_pick_active_node`, preferring one reporting status: “available”).
3. `storage_content()` and `storage_status()` per group storage.
4. `vm_config()` and `vm_snapshots()` per considered VM. A VM is *not* fetched at all when `exclude.running_only` is set and it is stopped — the one case `cluster/resources`'s own fields can decide without a config fetch. Every other VM is fetched, including config-excluded ones: whether a specific disk turns out to be “ungrouped” is only knowable after seeing which storage it is actually on, and a config-excluded VM's disk still needs its size for (C4)/(C5) (previous section).

This is the one step that runs across a bounded thread pool (`config.proxmox.read_workers`, REVIEW.md P-02): section 3.5's whole point is that `GET .../qemu/{vmid}/config` has no batch form, so for a several-hundred-VM cluster this is the read phase concurrency actually helps. It is split into `_fetch_vm()` (network I/O only, run by the pool, one call per considered VM) and `_join_vm_disks()` (pure — the same function the old sequential `_collect_vm_disks()` was renamed from, unchanged in what it computes). `ThreadPoolExecutor.map()` returns each VM's fetch in the same order `client.vm_resources()` listed it, even though the fetches themselves complete in whatever order the pool schedules them, so the join phase runs single-threaded and in the original order — `disks_by_group`, `referenced_volid`s and warnings end up byte-for-byte identical to a fully sequential run regardless of `read_workers` or of thread scheduling, and never need a lock. `PveClient`

itself may reauthenticate mid-fetch if several threads hit an expired ticket at once (50-pve-api.md's P-01 note); that reauthentication is what its own lock serializes, not this join.

5. Non-QEMU resources (type != "qemu") are skipped outright — this tool never touches LXC containers, and section 3.5's read/write paths are qemu-only throughout.

The lock field comes from `vm_config()`, not a separate call

Section 9.3's own pseudocode says `lock` is readable from either `/qemu/{vmid}/config` or `/status/current`. `_collect_vm_disks` reads it from the config response already being fetched for the disk join, so planning-time lock detection costs no extra API call. `/status/current` is still needed later, but only immediately before a specific move (section 9.2's pre-move re-validation, `execute.py`, not yet written) — never here.

Pin priority: `_pin_reason()`

One function, checked in the fixed order section 5.3 (C2) lists its conditions: config exclusion (VM-level, then `exclude.disks`), a real snapshot or an unreferenced companion volume (section 3.7, `_disk_snapshot_or_orphan_reason` — the "current" entry `GET .../snapshot` always returns, verified on a live cluster, is filtered out before counting), the per-disk cooldown (below), a VM lock, then an `unusedN` disk when `exclude.include_unused_disks` is false. A disk gets at most one reason; the first that applies wins, matching how an operator would explain it.

The per-disk cooldown pin

`build_topology()` takes an optional state: `state.State` and now: `datetime` (both default to "no cooldowns"/the real clock — see `docs/internals/15-state.md`) and computes `state.active_disk_cooldowns()` once per group, before the per-VM join loop, into `cooldowns_by_group: dict[str, dict[str, float]]` (bare `vmid:device` -> seconds remaining). Inside the loop, once a disk's `group_name` is known, `cooldowns_by_group[group_name].get(key, 0.0)` is passed to `_pin_reason()` as `cooldown_remaining_seconds` — a plain float, not a second `State` lookup inside that function, keeping `_pin_reason()` a pure decision over already-resolved flags exactly like every other condition it checks. The reported reason names how much of `gates.cooldown_per_disk` remains: `cooldown: moved recently, 1.0h left on gates.cooldown_per_disk`.

This pin is **not** exempted when the disk's storage is actively violating (C4)/(C5) — see `docs/internals/15-state.md`'s "Deliberately not implemented" section for why, and why that gap is bounded and safe rather than silently wrong.

Sizes: content is authoritative, config is the fallback

`_resolve_disk_size_and_format` looks up the volume in that storage's content listing first (section 3.5: authoritative for what is actually allocated) and falls back to parsing the VM config's own `size=` only when the volume is missing from content — logged as a warning naming the disk, since assume-thick-provisioning sizing from a stale or unlisted config value is a real, if rare, source of drift. Two independently verified reasons this fallback path is not merely defensive: the `Datastore.Allocate-vs-Audit` privilege gap (50-pve-api.md), and an `unusedN` volume, deleted directly on the storage backend outside Proxmox (confirmed with the cluster's operator), that PVE's own config still referenced and that PVE itself never noticed or refused to boot the VM over — `IMPLEMENTATION_PLAN.md` section 3.6's note. `_parse_pve_config_size_bytes` parses PVE's own config-file size suffixes (512G meaning binary GiB, no explicit i) — deliberately not `units.py`'s parser, which is for *this project's* config file, a different and coincidentally similarly-shaped format.

U_se_xt: everything not referenced

Every disk actually placed into a group's `D` records its valid in `referenced_valids[storage_id]` as it is processed. `_build_storages` then sums the size of every content entry on that storage whose valid was

never referenced — orphans, templates, other groups’ foreign volumes, and disks of VMs this run never fetched (stopped-and-excluded, or ungrouped) all fall out of this one computation with no special-casing per category, which is exactly section 5.1.1’s definition read literally.

reserve.py: one (C4)/(C5) evaluator, shared

`compute_reserve_status()` is deliberately its own module, not a method on `Storage`: the solver (`optimize.py/heuristic.py`, not yet written) will need to evaluate the identical formula against a *candidate* assignment, not only the current one, and `pve-storage-drs show-load`’s reporting needs it against the current assignment today. Both call the same function (`AGENTS.md` section 5) — `show-load`’s use of it is already exercised end-to-end (see `cli.py`’s `_render_show_load_human/_json`). The section 14 worked example’s initial state (`tests/unit/test_reserve.py`) is the proof this reproduces the plan’s own arithmetic exactly, not merely a self-consistent unit test.

The load model

What does this page answer? How does `loadmodel.py` turn six raw Prometheus series into the single per-disk ℓ_d section 4 defines, and what happens when the data for one disk cannot be trusted? Describes `proxmox_storage_drs/loadmodel.py`.

One function, one group, already-built input

`compute_group_load()` takes an already-built `Topology.Group` (one group’s D/S, section 5.1) and a `metrics.PrometheusClient` and returns a `GroupLoad`: one `DiskLoad` per disk (ℓ_d) and one `StorageLoad` per storage (L_s, u_s), plus the group’s u^* and whether the whole group is idle. It never touches the PVE API and never builds topology itself — matching `reserve.py`’s identical “given the join, evaluate one formula” framing, not a coincidence: both are meant to be called again, unchanged, by the solver once it exists.

Six queries, not six-times-groups queries

The six raw quantities (section 3.4) are fetched with **one instant query each**, unfiltered by group — `sum by (vmid, device) (...)` already returns every disk Prometheus currently reports, and picking out one group’s keys in Python is free, whereas six *more* Prometheus queries per group is not. `read_time_ns/write_time_ns` are converted from nanoseconds to seconds here, engine-side (`_combined_raw`), for the same reason F-03 moved `read_factor/write_factor` engine-side: one tested constant beats the same `/1e9` repeated across deployed PromQL strings.

Known limitation, not a bug: a config with more than one group re-runs these same six unfiltered queries once per group — correct (each group’s coverage/rejection decisions are independent) but wasteful for a many-group cluster. Fetching once and slicing per group would be a straightforward follow-up if a real deployment’s group count ever makes this Prometheus load worth avoiding; nothing about `compute_group_load()`’s per-group signature forces the current one-fetch-per-call shape, it is just what the first implementation does.

Coverage rejection excludes a disk from the group total, not just from its own ℓ_d

Section 3.4: “reject a disk whose sample coverage over W is below `window.min_coverage` and fall back to its last known load from `state.json`... never treat missing data as zero load.” Two things worth being explicit about, since the plan states the rule but not this mechanism:

1. **The rejected disk’s raw contribution is excluded from $T_g/0_g/B_g$ entirely**, not merely capped or zeroed in place — one noisy or half-missing series must not bias every *other* disk’s normalized share. ℓ_d for every accepted disk is computed from the accepted-only totals.

2. **The rejected disk’s own ℓ_d is substituted afterward**, from `last_known_loads` if the caller has one (keyed by `topology.Disk.key`), or flagged `0.0` if not. `DiskLoad.flagged_reason` is set either way, so a caller can never mistake a flagged `0.0` for a genuinely idle disk — `show-load` renders it as an explicit Δ line, never silently.

`state.py` (section 11.2) now provides the real `last_known_loads` source: `cli.py`’s `show-load` and `plan` both pass `state.load_vector_for_group(state, group.name)` through unchanged, exactly the mapping this function already expected (see `docs/internals/15-state.md`). `compute_group_load()` itself needed no change at all — the parameter was shaped for this from the start. A group with no recorded balance yet (a fresh `state.json`, or none on disk) still gets `None`, and every coverage-rejected disk with no seed is flagged `0.0`, exactly as before.

tpmstate0/unusedN are never rejected; efidisk0 is

Section 3.4’s own note: `tpmstate0` and `unusedN` are not QEMU block devices and legitimately emit no `blockstat` series — $\ell_d = 0$ for them is correct, not a data-quality problem. `_is_metrics_expected_absent()` encodes exactly this device-name distinction so these two kinds are never flagged for low coverage, while `efidisk0` — a real QEMU drive that *should* appear — is held to the same `min_coverage` bar as `scsi/virtio/ide/sata`. Getting this backwards either way is a real bug: exempting `efidisk0` would hide a genuine metrics gap on it; not exempting `tpmstate0/unusedN` would spam a flag on every run for every cluster, for a “gap” that isn’t one.

$T_g = 0$ is “idle”, computed only from accepted disks

`GroupLoad.idle` is `True` exactly when every *accepted* disk’s `raw_t` is zero — the guard section 4 requires (“if $T_g = 0$ the entire group is idle, skip it”). An all-rejected group (e.g. a total Prometheus outage mid-window) also comes out `idle=True` by this same computation, which is not quite literally accurate (the truth is “unknown”, not “idle”) but is harmless: both mean “do not act”, which is the only decision that follows from either. A caller that wants to distinguish the two can — every disk’s `flagged_reason` is still there to check.

L_s/u_s are the *current* assignment, not a candidate one

`StorageLoad` sums ℓ_d over disks whose `Disk.current_storage` already points at that storage — today’s state, exactly like `reserve.py`’s `compute_reserve_status()`. The solver (`optimize.py/heuristic.py`, not yet written) will evaluate the identical per-disk ℓ_d values against a *candidate* assignment instead; nothing in `loadmodel.py` needs to change for that, since ℓ_d itself does not depend on which storage a disk is currently on.

What is still missing from phase 3

`IMPLEMENTATION_PLAN.md` section 12 phase 3’s own “done when” is “correct act/no-act decision per group, with the reasoning shown” — that is the section 6 drift/imbalance gates, cooldowns, and the reserve override, none of which exist yet. `loadmodel.py` is the load computation those gates consume, not the gates themselves; `pve-storage-drs show-load` reports $\ell_d/L_s/u_s$ today, but no command yet decides whether a group should be re-balanced.

Gating: deciding whether to act at all

What does this page answer? How does `gates.py` turn a `loadmodel.GroupLoad` and per-storage `reserve.ReserveStatus` into the section 6 act/no-act verdict, and why isn’t cooldown handling here? Describes `proxmox_storage_drs/gates.py`.

One function, three gates, in the order section 6 lists them

`evaluate_group_gates()` is pure — no network I/O, no `state.json` access — taking a group's already-computed `GroupLoad`, a `storage id -> ReserveStatus` mapping (the caller's job, via `reserve.compute_reserve_status()` — this module never recomputes it, per AGENTS.md section 5's "one implementation of every rule"), the group's `GatesConfig`, and the load vector recorded at the group's last *executed* balance (`last_load`, `None` if there isn't one). It checks, in section 6's own order, stopping at the first that decides:

1. **Reserve override.** Any storage in `reserve_statuses` with `.violated` set forces `act=True` immediately, bypassing drift and imbalance entirely — section 13's "safety is not subject to hysteresis" made literal, not just documented intent.
2. **Drift gate.** $\|l_{\text{now}} - l_{\text{last}}\|_1 / \|l_{\text{last}}\|_1 \geq \text{gates.drift_threshold}$, vectors aligned over the **union** of disk keys (a disk absent from one side contributes its full load in the other as drift — a new or deleted disk is a genuine change to the group's I/O profile). `last_load=None` skips this gate outright, per section 6's own degenerate-case table — not "treat as zero drift", which would make an operator's very first run fail to act on an already-imbalanced cluster.
3. **Imbalance gate.** $(\max_s u_s - \min_s u_s) / u^* \geq \text{gates.imbalance_threshold}$, u^* from `GroupLoad.average_utilization`. $u^* == 0$ (idle group, section 4) always means no-act — there is nothing to spread evenly.

last_load is real now, for show-load and plan

`state.py` (section 11.2) exists: `cli.py`'s `_last_loads_by_group()` reads `state.path` once per run and hands each group's slice of `last_balance.load_vector` (re-keyed to plain `topology.Disk` key, or `None` if that group has no recorded balance yet) to `evaluate_group_gates()` as `last_load` — both `show-load` and `plan` do this identically (`cli.py` calls the one function, not two copies). `evaluate_group_gates()` itself needed no change at all: it already took `last_load` as a parameter and treated `None` correctly (see `docs/internals/15-state.md` for the read side, and this page's next section for what is still missing).

What this does not yet do. `state.json`'s `last_balance` is only ever *read* here — nothing in this codebase calls `state.with_recorded_balance()` yet, because nothing executes a migration yet (`execute.py`, phase 7). Until then, `last_balance.load_vector` for any group stays exactly what it was the last time a human (or a test) wrote it by hand; a config with no `state.json` on disk, or a fresh install, still behaves exactly as this page originally described — `last_load=None`, drift gate skipped, decision reduces to "reserve override, else imbalance". The wiring is real; the writer that would make it self-sustaining is not built yet.

show-load's gate line is diagnostic, not a real decision

`pve-storage-drs show-load` prints each group's verdict (`Group <name> → ACT: <reason> / → NO ACTION: <reason>`) reusing `GateDecision.reason` verbatim — deliberately the single source of truth for the wording, rather than a second, similarly-but-not-identically phrased string living in `cli.py`. Its verdict now reflects real drift history whenever `state.json` has one, exactly like `plan`'s does — but `show-load` still isn't what `apply` (not yet written) will actually gate an execution on; it is `plan/apply`'s own gate evaluation, run at the moment a plan is built or applied, that is authoritative.

Cooldowns are not this module's job

`gates.cooldown_per_disk/cooldown_per_storage` decide which *individual* disks or storages are movable once a plan is already being built — the same kind of decision (C2)'s other pin reasons make in `topology.py`, not a group-wide act/no-act verdict, so this module still has no cooldown logic of its own. Both are now implemented, in the modules this page always said they belonged with: the per-disk cooldown is a `topology.py` (C2) pin (`docs/internals/60-topology.md`), and the per-storage cooldown is a `heuristic.py` target-eligibility filter (`docs/internals/90-heuristic.md`), both reading `state.py`'s cooldown data (`docs/internals/15-state.md`).

The heuristic solver

What does this page answer? How does `heuristic.py` turn a group’s loads into a target assignment, why is the section 5.4 objective its own function rather than baked into the search, and what does this module deliberately not do yet? Describes `proxmox_storage_drs/heuristic.py`.

`evaluate_assignment()` is the objective, kept separate from the search

Section 5.5 requires it explicitly: “the heuristic must use the **same** feasibility and objective functions as the MILP path so the two backends are directly comparable.” `evaluate_assignment()` takes a Group and an arbitrary candidate Assignment (`dict[Disk.key, storage id]`, not necessarily reachable from the current state by any sequence of moves) and returns an `ObjectiveBreakdown` — the section 5.4 objective’s five terms kept separate, not collapsed into a single number, because `explain` (not yet written) needs the arithmetic shown, and because the tests cross-checking this against section 14’s worked example need to assert each term individually, not just a total that could be right for the wrong reasons.

This is the same design choice `reserve.py` made first: a pure function of *data*, with no notion of “the current run” baked in. `heuristic.py` reuses `reserve.compute_reserve_status()` directly for (C4)/(C5), passing it a `storage_of` callback closed over the candidate assignment being evaluated — the extension point `reserve.py`’s own docstring already anticipated (“the solver will pass a candidate assignment through the identical shape”) before this module existed to use it.

Repair is unconditional, not weight-driven

Section 13: “the reserve is never traded against balance.” The heuristic makes this literal rather than merely well-weighted: `_repair()` runs *before* `_descend()` and fixes any (C5) violation by moving disks off the worst-violating storage, one iteration at a time, regardless of what the objective weights say — there is no β/γ value large enough to switch this off. `objective.reserve_violation_penalty` still appears in `ObjectiveBreakdown.reserve_penalty_term`, but only for *reporting* a residual that repair could not fix (physically impossible, not merely unattractive) — by the time `_descend()` runs, every reserve violation `_repair()` could resolve already has been, so in the common case this term is exactly zero and does not influence `descend`’s choices at all. `test_reserve_violation_is_repaired_even_with_beta_high_enough_to_forbid_balance_moves` is the regression test for this: $\beta=100$ (so no purely-cosmetic balance move could ever be worth a migration) still sees the repair move happen.

A candidate is judged by the group-wide total shortfall, not the source storage’s own. Moving a disk off a violating storage always reduces *that* storage’s own shortfall — it has fewer bytes, and if anything a smaller largest-disk requirement — which makes “does the source’s shortfall fall” a tempting but wrong test: it says yes even when the disk only relocates the problem to whichever storage receives it, or makes the group’s total worse. An earlier version of `_repair` used exactly that test and would oscillate a disk back and forth between two storages neither of which can fully hold it, “successfully” reducing the current worst offender’s shortfall on every iteration while never making net progress. Requiring the *group* total to strictly decrease fixes this and is also what makes the loop’s iteration bound (movable disks times storages, generously sized rather than tightly) actually a correct termination argument rather than a lucky one. `test_repair_does_not_oscillate_when_no_target_can_fully_absorb_the_violation` is the regression test: a 3 TiB disk that cannot fit, alone, on either of two 8 TiB/`reserve_factor`: 2.0 storages is moved exactly once (repairing what can be repaired, from a 3 TiB group-wide shortfall down to 1 TiB), not shuffled back and forth forever.

Descend explores swaps, not only single moves

Section 5.5 is explicit that swaps are not an optional refinement: “when both storages are near their capacity limit, no single move is feasible, and only an exchange of two disks can improve the balance.” `_descend()` evaluates every single-disk move *and* every pairwise swap of two movable disks each iteration,

applying whichever most reduces the objective, and stopping when nothing does or after heuristic iterations. Both are plain re-evaluations of `evaluate_assignment()` against a trial dictionary — no separate swap-feasibility logic exists, because a swapped assignment is just another candidate the same objective function scores. `test_descend_uses_a_swap_when_no_single_move_is_feasible` constructs exactly the capacity-locked scenario the plan describes (two storages each with headroom smaller than any single disk) and checks the only reachable improvement — a swap — is the one found.

The storage cooldown excludes a destination, never a source

Section 6: “a storage involved in a migration within `cooldown_per_storage` accepts no new incoming moves.” `run_heuristic()`’s `cooldown_storages` parameter — a plain `frozenset[str]` its caller (`cli.py`, via `state.active_storage_cooldowns()`) computes, not a `state.State` this module reads itself — is threaded through to `_descend()` only. Both of `_descend()`’s neighbourhoods respect it: the single-move loop skips any `target.id` in `cooldown_storages`, and the swap loop skips a pair whenever *either* storage a disk would land on is in cooldown (a swap always sends one disk to each of the two storages it touches, so either side landing on a cooldown storage blocks the whole swap). A disk already resident on a cooldown storage is always free to move *away* from it — the rule is about new arrivals, not existing occupants — which is why the filter only ever appears on the destination side of a candidate, never the source.

`_repair()` never receives `cooldown_storages` at all — a deliberate exemption, not an oversight, matching this module’s own repair-is -unconditional stance above: a storage actively needed to resolve a live (C4)/(C5) violation is not deferred for having been written to recently, the same reserve-override principle `gates.py` and `payback.py` already apply to the drift/imbalance gates and the payback test respectively. `test_run_heuristic_repair_ignores_storage_cooldown` is the regression test: a violation whose only viable full repair target is in `cooldown_storages` is still repaired. `test_descend_blocks_new_arrivals_onto_a_cooldown_storage` and `test_descend_still_allows_a_disk_to_move_away_from_a_cooldown_storage` cover `_descend()`’s own two sides of the rule, both against the section 14 fixture.

objective.spread_metric: two different quantities, not one rescaled

Section 5.4/(C6) offers two imbalance forms, and `evaluate_assignment()` implements both: “l1” (default) is $\alpha * \sum(e_s)$, every storage’s absolute deviation from u^* summed; “minmax” is $\alpha * \max(u_s)$ — the plan’s own $t \geq u_s$ for all s , the raw utilization of the single hottest storage. These are not the same quantity with a different aggregation: $\max(u_s)$ and $\max(e_s)$ can disagree, because a storage sitting far below u^* produces a large deviation e_s that minmax was never meant to notice (the manual’s own words: “indifferent to a second nearly-as-bad storage” — nearly-as-bad on the *high* side; a cold storage is not what minmax was designed to react to at all). `ObjectiveBreakdown` carries both `spread_e` (deviations) and `utilization` (raw u_s) regardless of which metric is active, computed once in the same loop, so switching metrics never means computing something the other mode didn’t already have on hand.

REVIEW.md’s Q-01 found this config knob silently ignored by an earlier version of this module (always L1, `objective.spread_metric` never read) — the same class of gap P-01/P-02 found in `pve.py/topology.py`. `test_spread_metric_minmax_is_indifferent_to_a_second_nearly_as_bad_storage` is the regression test, built directly from the manual’s own claim: two scenarios with an identical hottest storage but a materially different second-worst one score identically under minmax and differently under l1.

Proof this reproduces the plan, not just itself

`tests/unit/test_heuristic.py` doesn’t stop at “the assignment matches the prose.” At the default weights it asserts `ObjectiveBreakdown.total` equals 2.533333 (three-move plan) and, at `beta_move_count: 0.50`, 3.158333 (two-move plan) — the exact totals REVIEW.md Appendix A independently re-derived by hand from the plan’s own numbers, not values this module invented and then asserted against itself. Getting both to five decimal places is strong evidence the objective’s five terms, their units (TiB for

size, average in-flight I/O for load), and the search that picks among them are all correct together, not merely internally consistent.

What this pass deliberately does not do

- **“Polish” (heuristic step 4)** — reuniting a fragmented VM when doing so does not worsen imbalance beyond `gates.imbalance_threshold`. Not implemented: the section 14 fixture’s exact three-move and two-move solutions are both reachable by repair+descend alone (proven by the tests above), because the objective’s own kappa term already makes descend prefer co-location whenever it is not too costly. Polish would matter for an affinity fix that needs three or more disks to rotate simultaneously — outside single-move/pairwise-swap reach — which the one fixture that exists to validate this module does not exercise. A real gap, tracked here rather than silently absent.
- **(C2) format-compatibility eligibility** — `topology.Storage` does not yet carry the storage type/format information that rule needs (it is resolved internally in `topology.py`’s `_default_format` but never exposed on `Storage`), so every group storage is currently an eligible target for every movable disk. The section 14 fixture is homogeneous (three identically-typed storages) and does not exercise this gap either.
- **CP-SAT/CBC coefficient scaling (section 5.5)** — belongs to `optimize.py`, not yet written. `evaluate_assignment()`’s plain floating-point objective is what the MILP path will need to agree with, not a stand-in for its own separately-scaled integer objective. `heuristic.py` is wired into `cli.py`’s plan command, together with `schedule.py` (section 8’s move ordering) — see 95-schedule.md and docs/manual/27-plan.md. `explain/apply` remain stubs (docs/manual/30-safety-and-status.md).

The MILP backends: CP-SAT and CBC

What does this page answer? How does `optimize.py` turn section 5.3’s constraint set into a real CP-SAT/CBC model, why is the reserve solved in two lexicographic stages rather than one big-M objective, and what does `cli.py`’s backend dispatch actually do when a MILP solver is unavailable or fails? Describes `proxmox_storage_drs/optimize.py`.

Both backends solve, then hand off to `heuristic.evaluate_assignment()`

`solve()` builds a model, extracts the winning `x_{d,s}` assignment, and immediately calls the *same* `heuristic.evaluate_assignment()` the heuristic backend uses to produce the reported `ObjectiveBreakdown` — one implementation of the section 5.4 objective, shared by every backend (AGENTS.md section 5; `heuristic.py`’s own module docstring was written anticipating exactly this: “built specifically so `optimize.py` can call the identical function”). A modeling mistake in this module can therefore make the solver choose a *suboptimal* assignment at worst — it can never make the tool *believe* an assignment is better than it is, since the number that ends up in `plan`’s output is never this module’s own objective value.

Lexicographic, not big-M — and why that is the only mode implemented

Section 5.3 describes two ways to keep a reserve violation from being “traded away” by a large enough balance gain: a single-stage objective with a calibrated penalty P (big-M), or a two-stage **lexicographic** solve — minimize $\sum_s r_s$ alone first, then fix that total as a hard constraint and minimize the real objective. `optimize.py` implements only the second, both because the plan itself says to (“use the lexicographic solve by default... needs no calibration, its correctness does not depend on T_g ”) and because there is no `solver.*` config knob that ever selects big-M at runtime — it exists in the plan text to be *compared against* the lexicographic solve, a comparison tests/fixtures/generate_expected.py already does by exhaustive enumeration on `reserve-tradeoff.yaml`, not something this module needs a second code

path for. $\sum r_s > 0$ in a lexicographic result provably means *no* assignment could do better, at any weight — never merely “unattractive”.

CP-SAT: section 5.5’s exact integer scaling

Every coefficient is folded and rounded exactly as section 5.5 specifies:

- `_LOAD_SCALE` ($K = 10^6$) scales every load-valued quantity (e_s, t, u^*). (C6)’s per-(disk, storage) coefficient is $a_{\{d,s\}} = \text{round}(K \cdot \ell_d / c_s)$ — folded and rounded *once*, per the plan’s own warning against computing $\text{round}(K/c_s)$ and multiplying separately (which rounds the capability weight itself and would make CP-SAT and CBC disagree for a non-integer c_s).
- `_WEIGHT_SCALE` ($W = 10^4$) scales every objective weight ($\alpha/\beta/\gamma/\kappa$). The γ term folds z_d into its own coefficient ($\text{round}(\gamma \cdot W \cdot K \cdot z_d^{\text{TiB}})$) rather than factoring out a standalone $\gamma_{\text{scaled}} = \text{round}(\gamma \cdot K / 2^{20})$ — the plan’s own worked example of *why* that shortcut is wrong: at the default $\gamma = 0.05/\text{TiB}$, it rounds to zero and CP-SAT would silently stop caring about disk size when choosing what to move.
- Every size-valued quantity ($Z_s, R_s, r_s, z_d, C_s, U^{\text{sect}}, \text{min_free_bytes}$) is a whole-MiB integer (`_mib()`), already exact, needing no scale of its own.
- `_RESERVE_FACTOR_SCALE` is this module’s own addition, not named in the plan: (C5)’s $R_s \geq f_s \cdot Z_s$ multiplies a *variable* (Z_s) by `reserve_factor`, which section 5.5’s “fold constants into a coefficient” recipe does not directly cover (that recipe is for a constant multiplying a *binary* decision variable). Expressed instead as $\text{SCALE} \cdot R_s \geq \text{round}(f_s \cdot \text{SCALE}) \cdot Z_s$ with $\text{SCALE} = 10^6$ — a single linear constraint, correct to within $1/\text{SCALE}$ MiB, utterly negligible next to any real disk size.

`AddHint(x[d,  $\sigma_0(d)$ ], 1)` warm-starts every candidate from the current assignment, in both stages (harmless when unused, and stage 1’s own optimum is frequently “keep everyone where they are” when nothing already violates (C5)).

CBC: “direct transcription”, deliberately not scaled

The plan’s own words for this backend: “continuous e_s, Z_s, r_s are fine.” `_cbc_feasibility_constraints()/_cbc_objective_terms()` write the identical constraints CP-SAT does, but with plain floats and PuLP’s `prob.add_variable(...)` (not the older `pulp.LpVariable(...)` constructor PuLP’s own v4 migration deprecates) — no `_LOAD_SCALE/_WEIGHT_SCALE` anywhere, because CBC needs none of CP-SAT’s integer-coefficient discipline. `solver.mip_gap` and `solver.time_limit_seconds` map onto `pulp.COIN_CMD(gapRel=..., timeLimit=...)` directly.

(C1)/(C3)/(C4)/(C5) are shared code, factored once per backend

`_cpsat_feasibility_constraints()/_cbc_feasibility_constraints()` build exactly the constraints both lexicographic stages need identically — only the *objective* differs between stage 1 (`Minimize( $\sum r_s$ )`) and stage 2 (the real section 5.4 objective, with $\sum r_s$ fixed to stage 1’s result as a hard constraint). Each is called twice, with fresh variables each time, rather than trying to mutate one model’s objective in place — simpler to reason about than partial reuse, at the cost of building the constraint set twice per solve (irrelevant next to a MILP solve’s own running time).

(C3)’s $y_{\{v,s\}}$ linking honors `objective.affinity_counts_pinned_disks` exactly as the plan’s own text describes it: **false** (default) links y to x over D^{mov} only, so a pinned disk never forces $y_{\{v,s\}}=1$ for its storage; **true** ranges the constraint over all of D , which this module implements as forcing $y_{\{v,s\}}=1$ outright wherever a pinned disk of v already sits (the natural reading of “ $x_{\{d,s\}} \leq y_{\{v,s\}}$ for $d \in D$ ” when $x_{\{d, \sigma_0(d)\}}$ is a constant 1, not a variable, for a pinned disk).

cli.py's dispatch: auto cascades, an explicit backend still falls back

`cli._solve_group()` implements `solver.backend`'s four values: "auto" tries `cpsat` then `cbc`; "cpsat"/"cbc" try only that one; "heuristic" skips `optimize.solve()` entirely. Whichever backend `solve()` cannot use (library not importable, or no feasible solution within `solver.time_limit_seconds`) returns `None` — never an exception — and `_solve_group()` moves to the next one in the cascade, ending at `heuristic.run_heuristic()` if every MILP attempt failed. This applies **even to an explicitly forced backend**: section 13's failure-mode table says "solver infeasible or timing out -> fall back to the heuristic; never emit a partial/unvalidated assignment", with no carve-out for `solver.backend: cpsat` specifically. The one difference: falling back from an *explicit* backend logs a warning (an operator who named `cpsat` to reproduce a specific result should not have to diff --json output to notice auto quietly ran instead); falling back within auto itself is expected and silent. `plan`'s human and --json output both report which backend actually produced each group's plan (`solver: cpsat (optimal) / "solver_backend": "cpsat", "solver_status": "optimal"`), precisely so that distinction is never hidden.

Deliberately not implemented in this pass

- **The storage cooldown is not enforced by either MILP model.** `solve()` accepts `cooldown_storages` for interface symmetry with `heuristic.run_heuristic()`, but hard-fixing $x_{\{d,s\}}=0$ for a cooldown storage inside stage 2 — after stage 1 has already fixed the *value* of $\sum r_s$, not the specific assignment that achieves it — can make stage 2 infeasible in a case the heuristic's sequential repair-then-descend never hits, because descend only ever *extends* repair's assignment rather than jointly re-deriving it against a competing hard constraint. `solve()` logs a warning whenever it is called with a non-empty `cooldown_storages`, so this is loud, not silent, the day `state.json`'s cooldowns stop being permanently empty (nothing calls `state.with_recorded_cooldown()` yet — see `docs/internals/15-state.md`).
- **(C2) format-compatibility eligibility** — the same gap `heuristic.py` already documents (`topology.Storage` does not expose storage type/format).
- **A live cluster's numpy/pandas footprint.** `ortools` pulls in `numpy` (and `numpy` pulls in nothing further this project cares about) purely as its own transitive dependency; nothing in this project imports `numpy` directly. A recent `numpy`'s bundled type stubs use syntax newer than this project's `python_version = "3.11"` mypy target, which mypy's `follow_imports = "skip"` (used for `ortools.*` itself) cannot rescue, because a stub-level *parse* error happens before mypy ever gets to apply a per-module setting to it. This surfaces only if `ortools` (via `pip install -e .[solver]`) or `statsmodels`'s own `[forecast]` extra happens to pull in an affected `numpy` version into the *same* venv `make typecheck` runs against — `make install`'s own `.[dev]` never does, so it does not affect this project's actual gate. Both MILP backends were still verified for real during development (`pip install -e .[solver]`, all of `test_optimize.py` passing and reproducing the section 14 and `reserve-tradeoff.yaml` fixtures' exact numbers for both CP-SAT and CBC) — see `tests/unit/test_optimize.py`'s own docstring.

Ordering the moves

What does this page answer? How does `schedule.py` turn a target assignment into an ordered, transient-feasible sequence of moves, and what happens when no order exists? Describes `proxmox_storage_drs/schedule.py`.

One loop, reusing heuristic.py's objective for the ratio

`order_moves()` implements section 8.2's pseudocode directly: while there are pending moves, find the ones that are transient-feasible right now, schedule the best by imbalance-reduction-per-byte (moves that resolve a currently-violating storage win outright, regardless of that ratio), apply it, repeat. "Imbalance reduction" is computed by calling `heuristic.evaluate_assignment()` twice (before/after the candidate

move) and diffing `.imbalance_term` — not a second formula, the same one `heuristic.py` already computes, so a plan's ordering agrees with whatever `objective.spread_metric` is configured exactly the way the assignment that produced it did.

cost_m is z_d — and that is not an approximation today

Section 8.2's ratio is “imbalance reduction / cost_m”. `cost_m` here is simply the disk's size in bytes, not `payback.py`'s real duration-based cost (mirror time plus, when `saferemove` is on, wipe time — and `wipe_throughput` is a per-storage config value, so two candidate moves off different sources can have genuinely different wipe costs even for the same `z_d`). `migration.bwlimit_bytes_per_sec` is one global config value, so every move's *mirror* duration alone is `z_d / bwlimit`, a constant divided into every disk's size equally — dividing an imbalance reduction by `z_d` and dividing it by `z_d / bwlimit` produce the **same ordering** (`bwlimit` is a positive constant common to every candidate). So `cost_m = z_d` reproduces the true mirror-only ordering exactly, but is an approximation once a group has moves whose `saferemove` wipe cost differs enough to change the ranking `payback.py`'s full cost would produce. Scoring candidates by `payback.compute_move_cost()` instead of `z_d` is a natural follow-up, not yet done: this module predates `payback.py` and has not been revisited to consume it (REVIEW.md R-04).

The transient invariant, called with a single-move set always

Section 8.1 defines a generalized invariant over a whole in-flight set `M` specifically so it can be “implement[ed] as the single feasibility predicate and call[ed] with `M = {m}` for the sequential case” — this module is that sequential case. `transient_invariant_ok()` checks one move against state, the assignment as of immediately before that move starts; every previously-scheduled move is assumed **fully complete** by then (mirror finished, and any `saferemove` wipe finished, source genuinely freed) before this one begins. That is the correct model for `execution.max_concurrent_migrations: 1` (the default, and the plan's own “recommended configuration”) — see the next section for why it is deliberately not generalized to more than one in-flight move yet.

The `min_free_bytes` floor is folded into the transient check the same way (C5) folds it into the steady-state one (`max(f_b * max(Z_b, z_d), min_free_bytes)`) — the plan's own section 8.1 formula does not mention the floor, but there is no reason a storage's absolute minimum free space should stop applying just because a migration happens to be in flight.

The heuristic can accept a residual violation; the scheduler cannot execute one

`heuristic.py`'s repair step can leave a storage still violating (C5) if no reachable assignment can fully clear it — a genuine, sometimes unavoidable outcome the objective's `reserve_penalty_term` reports rather than hides (see 90-heuristic.md). When `order_moves()` is asked to schedule the move that leads to exactly that residual state, the transient check correctly says no: landing a disk on a target that still would not satisfy (C5) is not transient-feasible, full stop, and this module never treats “the heuristic already decided this was the best available” as a reason to relax that. The result is a deadlock report for that move (section 8.3 option 3, “report... never force a move that breaches the reserve”) — an honest “this cannot be safely executed”, not a false all-clear. See `test_plan_reports_a_deadlock_when_even_the_best_target_still_violates` in `tests/unit/test_cli.py` for the exact scenario reproduced end to end.

final_assignment is the schedule's own truth, not the heuristic's target

`ScheduleResult.final_assignment` is the assignment produced by actually applying order in sequence, starting from the current placement — every disk, not only the moved ones. When nothing deadlocks it is identical to the heuristic's target assignment, but when deadlocked is non-empty (fully or, more subtly, only partially) it is not: a caller reporting an “after” utilization/spread must evaluate `final_assignment`, not `heuristic.HeuristicResult.assignment`, or it reports numbers for moves that were never actually

scheduled (REVIEW.md R-02 — `cli.py`'s plan originally did exactly this, correct only when `order_moves()` scheduled every move a plan proposed).

What this pass deliberately does not do

- **Concurrent scheduling** (`execution.max_concurrent_migrations > 1`). `payback.py` now provides the move-duration estimates this would need, but reasoning correctly about overlapping in-flight windows — which pairs of moves can safely run together under the generalized section 8.1 invariant — is not implemented. This module schedules strictly sequentially regardless of the configured value; section 8.1's own generalized invariant is what a future version would evaluate against the real in-flight set instead of `{m}`.
- **Ordering priority 2** ("moves that free space a later move needs") and **staging** (section 8.3 option 1). Both are refinements over a plan that is already feasible and safe without them — priority 2 only affects how well a plan front-loads its value if interrupted, and staging only helps in a genuine pebble-motion deadlock. Omitting them can only make this module report a deadlock in a case a more sophisticated scheduler would have found a way through; it can never produce an order this module thinks is safe but is not.
- **Cooldowns**, for the same reason `gates.py` doesn't implement them yet: `state.json` (section 11.2), which would record the timestamps a real cooldown check needs, does not exist.
- **The section 7.3 saturation check** — belongs with `payback.py` (implemented; see 96-`payback.md` for why the check itself is deferred).

Wired into plan, not yet into apply

`cli.py`'s plan command calls `heuristic.run_heuristic()` then `schedule.order_moves()` for every group the gate says to act on, and renders the result (see `docs/manual/27-plan.md`). Nothing calls `schedule.py` from apply yet, because apply itself does not exist (`execute.py`, phase 7+) — plan only ever computes and prints.

Migration cost and the payback test

What does this page answer? How does `payback.py` turn a scheduled move into a cost, a plan into a benefit, and the two into an accept/reject verdict — and why does a reserve-fixing plan always pass? Describes `proxmox_storage_drs/payback.py`.

Cost and benefit are computed from data other modules already have

`compute_move_cost()` needs only a `schedule.ScheduledMove` and its source topology.Storage — no new fetch, no new state. `duration_mirror` is `z_d / migration.bwlimit_bytes_per_sec`; `duration_wipe` is `z_d / saferemove_throughput` when `migration.account_saferemove_wipe` and the source has `saferemove` on, else zero. `compute_benefit_load_seconds()` takes two `heuristic.ObjectiveBreakdown.imbalance_term` values — the pre-plan and post-plan `E` — and multiplies their difference by `migration.payback_horizon_seconds`; `heuristic.HeuristicResult` already carries both (`initial_breakdown/breakdown`), so `cli.py`'s plan handler passes them straight through.

headroom_src/headroom_dst: a plan formula this project cannot fill in

Section 7.1 writes `duration_mirror_d = z_d / min(bwlimit, headroom_src, headroom_dst)`. Neither `headroom_src` nor `headroom_dst` is defined anywhere else in `IMPLEMENTATION_PLAN.md`, and no config field or topology.Storage attribute represents a per-storage effective throughput ceiling distinct from the one global `migration.bwlimit_bytes_per_sec`. `compute_move_cost()` uses `z_d / bwlimit` only — exactly what the formula reduces to whenever neither storage's own throughput is the binding constraint, which is the case `bwlimit` exists to enforce in the first place. This is recorded as a known simplification of the plan's own underspecified formula, not silently worked around.

The reserve-override exemption

Section 7’s payback test weighs a move’s cost against the *balance* benefit it buys. That framing has an edge it does not name: a move resolving an active (C4)/(C5) violation is not optional the way a balance-driven move is, and its “benefit” under the section 7.2 formula can easily be zero or even structurally zero — moving the only loaded disk in a two-storage group between the two storages changes which one carries it without changing *E* at all, no matter how urgently the move is needed for capacity reasons. Rejecting that move on economic grounds would contradict section 13’s own “the reserve is never traded against balance,” which `gates.py` already treats as absolute for the drift/ imbalance `gates.evaluate_plan_payback()` applies the identical rule here: **a plan containing any move with `resolves_reserve_violation=True` always passes the aggregate ratio test**, regardless of the computed ratio. The hard per-move `max_single_move_duration` rule is not exempted this way — section 7.3 lists it as applying “regardless of the aggregate test” precisely because it is an operational limit (a mirror that takes that long has other costs an economic ratio does not capture), not an economic one.

`test_plan_json_output` and `test_plan_json_output_accepts_payback_when_saferemove_is_off` in `tests/unit/test_cli.py` exercise both halves of this end to end: the same reserve-driven, zero-benefit move is accepted when its duration is within the limit and rejected (correctly, via the hard rule, not the economic one) when a slow `saferemove` wipe pushes it over `max_single_move_duration`.

What this pass deliberately does not do

- **The 3-retry re-solve-with-doubled-beta/gamma loop** (section 7.3) on aggregate payback failure. A real UX refinement — it converges on the smaller subset of high-value moves rather than abandoning the run — but not a correctness requirement: reporting accept/reject plainly already satisfies “reject a plan whose cost outweighs its benefit.” Implementing the loop means re-invoking `heuristic.py/schedule.py` with adjusted weights from inside `cli.py`’s own orchestration, which is where it belongs when it is built, not half-done inside this module now.
- **Automatically dropping an individually-rejected move** and re-solving without it. Reported instead (`MoveCost.exceeds_max_duration`, `PaybackResult.rejected_moves`) — the same “report, never force” choice `schedule.py`’s deadlock reporting already makes for an unschedulable move.
- **The section 7.3 saturation-ceiling defer check** (`L_during(s) <= saturation_ceiling * N_s`). It needs the forecaster’s upper bound over a horizon equal to a specific move’s own duration, which `forecast.py`’s Forecaster protocol does not expose (it answers for `window.lookback` only); and no group in this project’s own dogfooding cluster has `storages[].saturation_load` set, which the plan itself says makes this check’s absence “fully supported... loses only this one advisory check.” `max_single_move_duration` (implemented) and the transient reserve invariant (`schedule.py`, already enforced before a move is ever scheduled) are the two bounds section 7.3 calls “always active”; this deferred one is explicitly the best-effort extra.

Wired into plan, not yet into apply

`cli.py`’s `_handle_plan()` computes a `PaybackResult` for every group the gate acts on, right after scheduling its moves, and both render functions show it (`docs/manual/27-plan.md`). `compute_wipe_duration_seconds()` is also used by `verify-storages`’s existing wipe-time warning — one implementation of the formula, not two (AGENTS.md section 5). Nothing calls this module from `apply`, because `apply` does not exist yet (`execute.py`, phase 7+).